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Studierichting: Informatica
Oefeningenreeks 4A nr. 15.4

$$\int x^2 e^{4x} dx$$

$$\begin{cases} u = x^2 \\ dv = e^{4x} dx \end{cases} \Rightarrow \begin{cases} du = 2x dx \\ v = \dfrac{e^{4x}}{4} \end{cases}$$

$$= \frac{x^2 e^{4x}}{4} - \frac{2}{4} \int x e^{4x} dx$$

$$\begin{cases} u = x \\ dv = e^{4x} dx \end{cases} \Rightarrow \begin{cases} du = dx \\ v = \dfrac{e^{4x}}{4} \end{cases}$$

$$= \frac{x^2 e^{4x}}{4} - \frac{1}{2} \left[\frac{x e^{4x}}{4} - \frac{1}{4} \int e^{4x} dx \right]$$

$$= \frac{x^2 e^{4x}}{4} - \frac{x e^{4x}}{8} + \frac{1}{8} \left(\frac{1}{4} e^{4x} \right) + C$$

$$= \frac{x^2 e^{4x}}{4} - \frac{x e^{4x}}{8} + \frac{e^{4x}}{32} + C$$

References